
1 Sequencer Analysis

Summary

Menu	Production facilities management → Status analyses, machine data
Transaction code	psa
Function authorization	psa

Usage

The sequencer monitoring is used to monitor machine programs that are programmed in sequencers; in this way it is easy to determine if programs, e.g. cleaning, have not run without errors. A machine or aggregate can have several sequencers that run in parallel.

A sequencer consists of

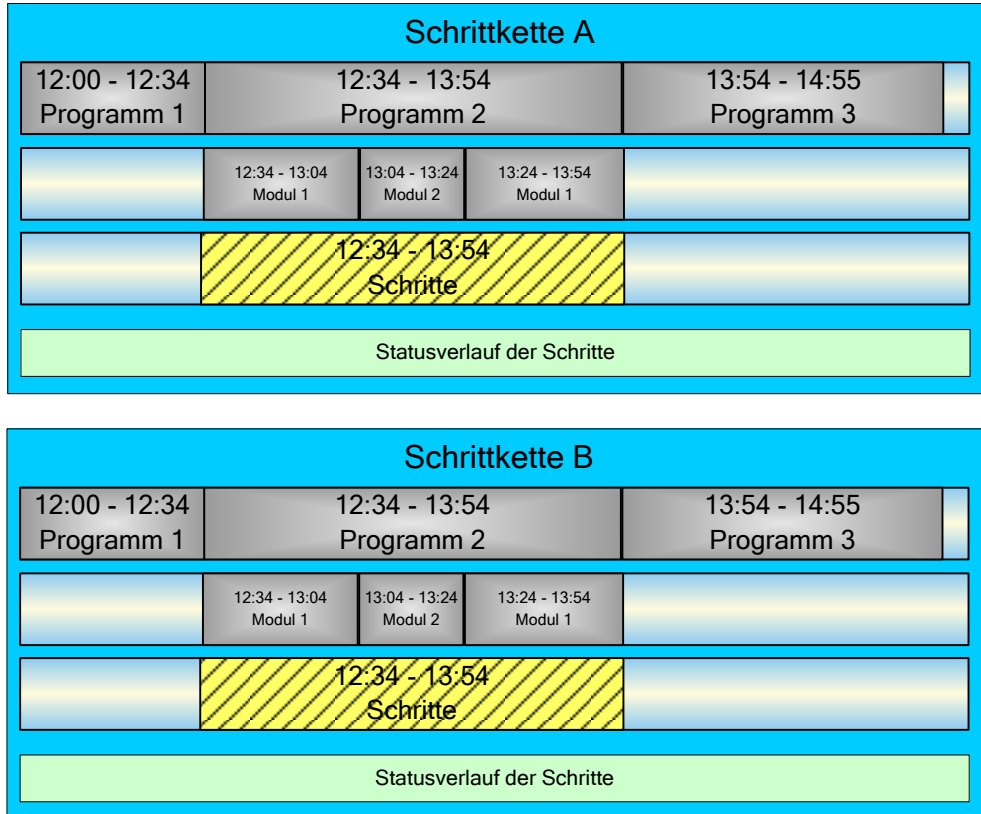
- Programs
- Modules
- Steps:

A machine can have multiple, independent, parallel sequencers. These are then defined by further TAGs.

The evaluation only considers data within the related evaluation period. In this way it cannot be determined which program was most recently active before the evaluation period. Due to this fact, the starting point of the evaluation must be selected such that at least one posting of the program to be evaluated is present after the starting point. Otherwise, the first program displayed is that which has its first posting after the evaluation start.

Example:

Schrittketten einer Maschine



Integration

Due to the quantity of data, sequencers are recorded as process values, but are interpreted for the evaluation as having parallel status. For this reason, both the parallel status and the PRD collection must be configured accordingly. The configuration is explained in the separate documentation for setting up the sequencer collection by a specific project.

Prerequisite

The sequencers are based on status words that are present in a control for the sequencers. These must be present on the machine.

Selection parameters

The following selection criteria are available in the application:

Machine

Selection of a machine for which the data is to be determined and displayed.

Period

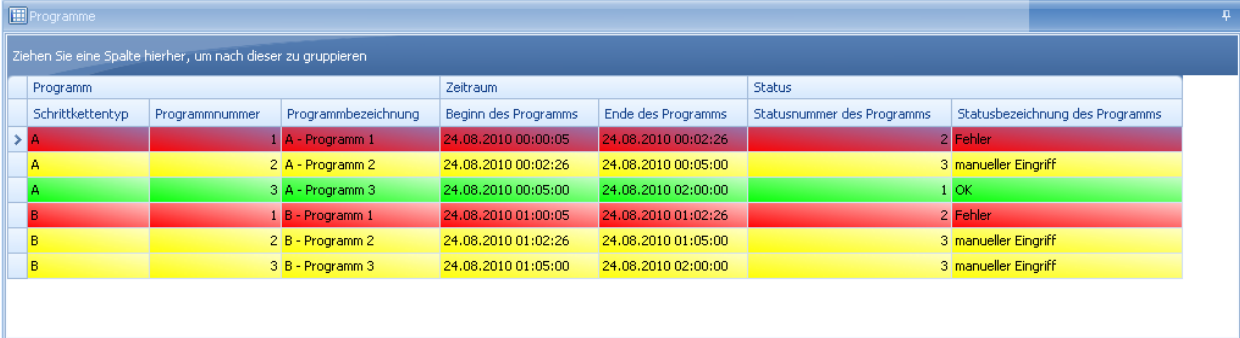
Period for which the data is determined and then displayed.

Consider long-term data

This flag can be used so that even data that has already been archived can be accessed.

Detail application programs

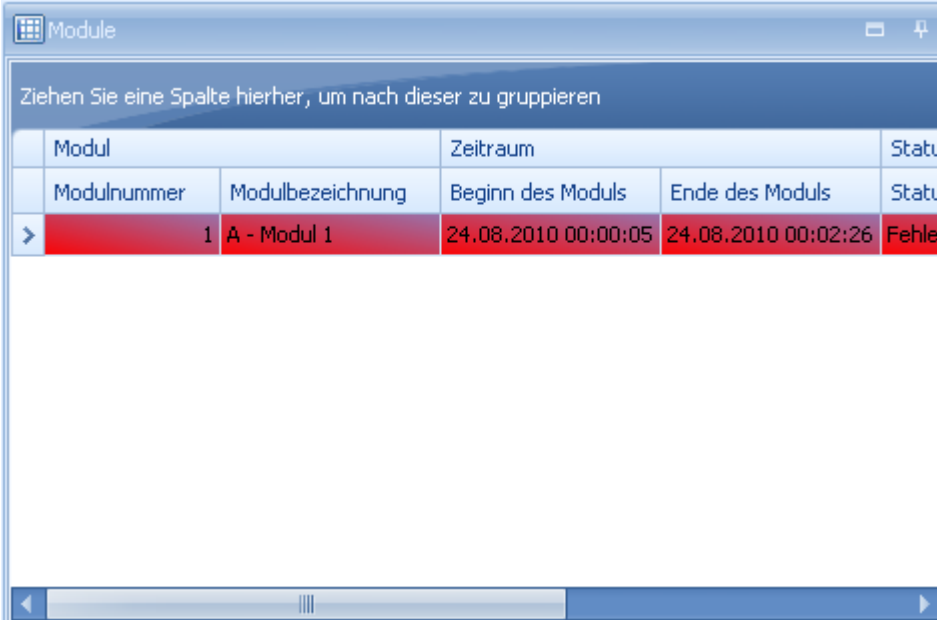
The program is the machine program that is currently running; do not confuse this program with the "program" of the Weihenstephan standard status model (#00200); instead it is the KHS machine program.



Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren						
Programm			Zeitraum		Status	
Schrittketten-typ	Programmnummer	Programmbezeichnung	Beginn des Programms	Ende des Programms	Statusnummer des Programms	Statusbezeichnung des Programms
> A	1	A - Programm 1	24.08.2010 00:00:05	24.08.2010 00:02:26	2	Fehler
A	2	A - Programm 2	24.08.2010 00:02:26	24.08.2010 00:05:00	3	manueller Eingriff
A	3	A - Programm 3	24.08.2010 00:05:00	24.08.2010 02:00:00	1	OK
B	1	B - Programm 1	24.08.2010 01:00:05	24.08.2010 01:02:26	2	Fehler
B	2	B - Programm 2	24.08.2010 01:02:26	24.08.2010 01:05:00	3	manueller Eingriff
B	3	B - Programm 3	24.08.2010 01:05:00	24.08.2010 02:00:00	3	manueller Eingriff

Detail application module

A module consists of a defined sequence of steps. Each module fulfills a certain task, which can be used multiple times within a program in another context. For example, the module "Fill boiler" can be used for filling with either water or cleaning medium.



Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren				
Modul		Zeitraum		Statu
Modulnummer	Modulbezeichnung	Beginn des Moduls	Ende des Moduls	Statu
> 1	A - Modul 1	24.08.2010 00:00:05	24.08.2010 00:02:26	Fehle

Note on determining the first module for the selected program: Using the starting point of the program, the data are determined minus one minute and the end point of the program due to the inaccuracy of the measurement. In the end, the first data record selected is one with an entry time that is less than or equal to the start of the program. If the point in time of this entry lies before the start of the program, it is corrected to the start of the program. If the start of the entry lies after the start of the program, another entry is inserted before it, which assumes the starting point of the program and has the value 99 (corresponds with "Module unknown").

The last entry contains the ending point of the program as the ending point.

Detail application steps

Each step has a certain, defined task. For each step there is a uniquely defined machine function (with a defined location of all process valves). Switching from one step to the next can be connected to various criteria.



Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren				
Schritt		Zeitraum		Status
Schrittnummer	Schrittbezeichnung	Beginn des Schritts	Ende des Schritts	Statusnummer de:
1	A - Schritt 1	24.08.2010 00:00:05	24.08.2010 00:01:01	
2	A - Schritt 2	24.08.2010 00:01:01	24.08.2010 00:02:01	
3	A - Schritt 3	24.08.2010 00:02:01	24.08.2010 00:02:23	
1	A - Schritt 1	24.08.2010 00:02:23	24.08.2010 00:02:26	

Note on determining the first step for the selected module: This procedure with the steps is analogous to the determination of the modules.